

EE2080: Practice Problems - Set II

1. The price X_n of a stock at the close of day n is modeled as a Gaussian random walk,

$$X_n = X_{n-1} + Z_n,$$

where the increments (Z_n) are i.i.d with $Z_n \sim \mathcal{N}(\mu, \sigma^2)$. The initial price is $X_0 = 42.3$.

- (a) Calculate the mean and variance of the stock price at the close of day 5.
(b) Give a 95% prediction interval for the stock price at the close of day 5. You may use the fact that if $W \sim \mathcal{N}(0, 1)$, then

$$P(W \leq 1.96) = 0.975.$$

- (c) After day 4, the prices at the end of each of the first four days have been observed to be

$$X_1 = 44.4, \quad X_2 = 44.0, \quad X_3 = 47.1, \quad X_4 = 47.8.$$

Update your prediction interval for the stock price at the close of day 5, and comment on how it differs from the prediction interval obtained in part (b).

2. A gambler models her total winnings as a simple random walk $\{X_n\}_{n \geq 0}$ with $X_0 = 0$, where, at each step,

$$X_n - X_{n-1} = \begin{cases} +1, & \text{with probability } p, \\ -1, & \text{with probability } 1 - p, \end{cases}$$

independently of all previous steps. The parameter p is unknown.

The first ten observations are $(X_1, \dots, X_{10}) = (1, 2, 1, 2, 3, 4, 5, 6, 5, 6)$.

- (a) Based on these observations, estimate the value of p .
(b) More generally, suppose you observe

$$X_0 = 0, \quad X_1 = x_1, \quad X_2 = x_2, \quad \dots, \quad X_n = x_n.$$

Give a natural estimator of p based on these observations.

- (c) The *maximum likelihood estimate (MLE)* of p is defined as the value

$$\hat{p} = \arg \max_{0 \leq p \leq 1} P(X_1 = x_1, \dots, X_n = x_n),$$

What is the maximum likelihood estimate of p ?

3. Let $\{X_n\}_{n \geq 0}$ be a simple random walk starting from $X_0 = 0$, where, at each step, the walk moves up by one with probability p and down by one with probability $q = 1 - p$. Find the following:

- (a) $P(X_6 = 2)$.
(b) $P(X_6 = 2 \mid X_3 = 1)$.
(c) $P(X_n = n - 4)$.
(d) $E[X_7]$.
(e) $E(X_9 \mid X_5 = 3)$.

4. Let $\{N(t), t \geq 0\}$ be the counting process of a Poisson process with rate $\lambda = 5$. Calculate the following:

- (a) $P(N(0.4) \leq 2)$.
(b) $E[N(6.4)]$.
(c) $P(N(0.5) = 0, N(1) = 1)$.

5. Suppose that telephone calls arrive at a call centre according to a Poisson process with rate $\lambda = 100$ calls per hour, and each call is answered independently with probability 0.6. What is the probability that there are no answered calls during the next minute?

[Note: Only concepts that have been covered in this course can be used without proof (specifically, do not use the concept of a thinned Poisson process OR you can prove it and then use it). Hint: You can use the Taylor series expansion of e^x to arrive at the final answer.]

6. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be two independent Poisson random variables.

(a) Show that

$$X + Y \sim \text{Poisson}(\lambda + \mu).$$

You may begin by writing

$$P(X + Y = z) = \sum_{x=0}^z P(X = x) P(Y = z - x).$$

(b) Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be independent Poisson processes with rates λ and μ , respectively. Use part (a) to show that the process

$$\{N_1(t) + N_2(t), t \geq 0\}$$

is a Poisson process with rate $\lambda + \mu$.

(c) Route 1 buses arrive at a bus stop according to a Poisson process with rate $\lambda_1 = 4$ buses per hour, while Route 6 buses arrive independently according to a Poisson process with rate $\lambda_6 = 2$ buses per hour. Suppose you have already been waiting at the bus stop for 5 minutes without seeing either bus. On average, how much longer do you expect to wait until the next Route 1 or Route 6 bus arrives?

7. Let $\{S_n\}_{n \in \mathbb{N}}$ be a Poisson process with rate λ . Here, S_n is the n^{th} arrival time. Let $\{N_t\}_{t \geq 0}$ be the associated counting process. Show that given $N_t = 1$, $S_1 \sim \text{unif}(0, t]$, i.e., prove that $P(S_1 \leq x | N_t = 1) = \frac{x}{t}$, for $0 \leq x \leq t$.

(Hint: The counting process has stationary and independent increments.)